The empirical recurrence for OEIS A237304: recovering a conjecture that is not written down

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Abstract

OEIS A237304 records "Empirical recurrence of order 87", with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 450$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 87, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 87 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry's recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A237304 is "Number of (n+1)X(4+1) 0..2 arrays with the upper median of every 2X2 subblock equal.". It has offset 1 and begins

$$11545, 731603, 36167196, 2009833839, 108105728913, 5924015749908, 323852801203185, 17797884221279895, \ldots$$

The entry states:

> Empirical recurrence of order 87 (see link above)

As of the "Last modified" line on the live entry (Jul 23 2025 10:02:41, revision 7) this is still recorded as empirical. The recurrence's coefficients are not in the entry text.

2 The count is a walk count

The entry's condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u, and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 450$ vertices, $a(n) = \iota^{\top} M^{j(n)} \tau$ for a linear reindexing j fixed by the entry's shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \ldots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' =$ 450.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 900$ exact terms of the model returns order 87 — the order the entry states — with the integer coefficients

$$a(n) \;=\; \sum_{i=1}^{87} c_i\, a(n-i),$$

c_1	86	c_{23}	63185412384541408256092925	c_{45}	−809354836256
c_2	975	c_{24}	15186161990624697163687382172	c_{46}	−162286515676
c_3	−201724	c_{25}	−11761663653485375521007612339	c_{47}	4133088696023
c_4	413404	c_{26}	−487165430289073289790726149837	c_{48}	6832674954907
c_5	187851766	c_{27}	594511029465960704517850064336	c_{49}	−1761560583753
c_6	−1158176897	c_{28}	12703285683457614836618318115870	c_{50}	−2403272727569
c_7	−94991628405	c_{29}	−19717524527767102565618650754862	c_{51}	62580691421824
c_8	811283460639	c_{30}	−271648197751561915676007588764260	c_{52}	70464201627437
c_9	29466304311922	c_{31}	489523047215539128242512714665312	c_{53}	−1849148844448
c_{10}	−307572677411991	c_{32}	4796709694756065242288459539958788	c_{54}	−1717652811183
c_{11}	−5962954573982010	c_{33}	−9555224262391769221476827336859812	c_{55}	45312387643714
c_{12}	73491983545241066	c_{34}	−70311312467947914608823636912211075	c_{56}	34701884231661
c_{13}	816234507760160370	c_{35}	150304913531311255760195580790174587	c_{57}	−9174008605612
c_{14}	−11899779636995822640	c_{36}	858952289841667146295626695783910868	c_{58}	−5789667755585
c_{15}	−77232245935017961335	c_{37}	−1932945801253936655306548437664932795	c_{59}	152765481777482
c_{16}	1371171378907015075585	c_{38}	−8769910768567988081008350576074463532	c_{60}	79433445722429
c_{17}	5065370109655321922996	c_{39}	20506931573502000241237908298986460406	c_{61}	−20808924109004
c_{18}	−116702457609923883029278	c_{40}	74970976637642829531087007743615900985	c_{62}	−8916918040636
c_{19}	−221112701741800981264591	c_{41}	−180520695352788339575718280993138372732	c_{63}	230364780670769
c_{20}	7553962350823937107470613	c_{42}	−537123242329301351575290991344943760985	c_{64}	81397147946456
c_{21}	5147979218092924430532598	c_{43}	1323291333580673757554764787390855768868	c_{65}	−20567059055641
c_{22}	−380492271562480089481168918	c_{44}	3225612533289449711356971339176333805768	c_{66}	−5995450595616

Theorem 3. *The recurrence above holds for every n, and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 87, possibly only from some index on. Let q be that recurrence's characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $87+4$ terms removed returns order 87 as well, so $\deg q_{\min} = 87 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 12 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry's English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $87+4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 87 that this sequence satisfies — and that family turns out to have exactly one member.

References

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